

On-Chain Parametric Cost Insurance for Underserved Latin-American Digital Workers

Record of Closed Iterations and Request for Methodological Guidance

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§1. Motivation, problem, suggested solution, challenges

Motivation. We study Latin-American digital workers paid in Colombian pesos who carry recurring USD-denominated professional-input obligations (subscription AI services, API usage fees, cloud compute, software-as-a-service spend), a population whose macro-risk exposure is dominated by FX volatility: for the Colombian case specifically, the FX market is the efficient aggregator of macro shocks, with TES yields and core inflation operating as followers of the COP/USD path (Rincón-Torres et al., 2021, 2023; Rincón-Castro et al., 2021; Fuentes et al., 2014); the same FX-as-primary-transmission pattern is documented across emerging markets generally (Calvo and Reinhart, 2002; Rey, 2015; Bruno and Shin, 2015).

Problem. When the COP weakens against the USD, the USD-denominated cost line translates into unbudgeted COP expenditure, and no Colombian retail-bank product currently hedges the resulting FX-pass-through cost shock at the hundred-to-low-thousands-USD monthly notional these workers carry.

Suggested solution. We identify the macro risks X that quantitatively produce these cost shocks, validate them empirically, and design parametric cost-insurance products that compensate for the FX-pass-through exposure (Carter et al., 2017; Mahul and Stutley, 2010; Clarke, 2016). The demand pattern we target — underserved-country households purchasing depreciation insurance in the form of a call option on the foreign-exchange rate — is documented in a recent practitioner interview (Leifke, 2026): “the instrument that people really care about, especially in emerging markets, is more of like a devaluation insurance policy ... call options are kind of perfect for this because you don’t need a credit relationship. You just pay a premium upfront.” The intermediation product compensates the worker in a unit she actually spends (COP, or a COP-pegged purchasing-parity stablecoin); the implementation venue is on-chain settlement infrastructure (described in the 37-page technical companion §4), not the economic primitive.

Challenges.

- Identification posture — predictive versus causal (see §4.2).
- Convex-payoff sufficiency under insurance-underwriting standard (see §4.2).
- Multi- Y differential calibration; familywise versus false-discovery correction across legs (see §4.2).
- Anti-fishing protocol defensibility — pre-registration thresholds $N_{\text{MIN}} = 75$, $\text{POWER}_{\text{MIN}} = 0.80$, $\text{MDES}_{\text{SD}} = 0.40$ (see §4.2).
- Cross-iteration familywise error and iteration-selection bias (see §4.2).

Bridge to §2. We have run five iterations to date, all closed or parked; the model specifications, results, and failure post-mortems follow in Section 2.

§2. Five iterations

Pair D — BPO offshoring through COP/USD lag (PASS). Colombian young workers (14–28) employed in the broad CIU Rev. 4 services aggregate (Sections G–T), with Y_t the logit of the monthly young-worker share on DANE GEIH micro-data and X_{t-k} the lagged log Banrep TRM end-of-month spot.

$$Y_t = \alpha + \beta_6 X_{t-6} + \beta_9 X_{t-9} + \beta_{12} X_{t-12} + \varepsilon_t$$

where $Y_t = \text{logit}(\text{share}_t^{\text{GT}})$ and $X_{t-k} = \log(\text{COP/USD}_{t-k})$ for $k \in \{6, 9, 12\}$ months; the composite is $\hat{\beta}_{\text{composite}} = \hat{\beta}_6 + \hat{\beta}_9 + \hat{\beta}_{12}$. *Data.* **Source:** DANE GEIH micro-data (young-worker share by sector); Banrep TRM end-of-month spot. **Frequency:** monthly. **Window:** 2015-01-31 through 2026-02-28. **Sample:** $N = 134$ monthly. **Transformations:** logit on share- Y ; log-level on FX; lag panel $k \in \{6, 9, 12\}$; HAC($L = 12$) Newey-West SE; FEX_C_2018 expansion factors. **Repo:** [notebooks/bpo_offshoring_fx_lag](#). **Spec sha:** 964c62cca0...ef659. *Result.* $\hat{\beta}_{\text{composite}} = +0.13670985$, HAC SE 0.02465, $t = +5.5456$, one-sided $p \approx 1.46 \times 10^{-8}$; R1–R4 zero sign flips, classifier AGREE. *Interpretation.* PASS — gate cleared on lagged-FX predictor. *Caveats inherited from independent review.* The estimate hedges a correlation, not a causal identification of the BPO transmission channel; the lag effect is concentrated at the six-month horizon ($\hat{\beta}_6 \approx +0.109$, about 80% of the composite) rather than spread across the 6–12-month window; the post-2014 window over-represents the oil-collapse, COVID, and 2022–2024 Fed-tightening regimes; the verdict is sensitive to a single off-spec design choice (an orchestrator-brief variant including a `marco2018_dummy` in the primary returns $\hat{\beta}_{\text{composite}} = +0.0815$ at $p \approx 0.080$); and the dev-AI Stage-1 R2 Section-M reframe (below) suggests the operative sub-aggregate is consultancy/scientific/administrative services rather than Section-J ICT-narrow tech offshoring.

dev-AI Stage-1 — Section J ICT through COP/USD lag (FAIL). Colombian young workers (14–28) restricted to CIU Rev. 4 Section J (Information and Communication, Divisions

58–63), a strict subset of the Pair D cut, with the same monthly cadence, FEX_C_2018 expansion factors, and lag panel.

$$Y_t = \alpha + \beta_6 X_{t-6} + \beta_9 X_{t-9} + \beta_{12} X_{t-12} + \varepsilon_t$$

where $Y_t = \text{logit}(\text{share}_t^J)$ and $X_{t-k} = \log(\text{COP}/\text{USD}_{t-k})$; pre-registered $H_1 : \beta_{\text{composite}} > 0$. **Data. Source:** DANE GEIH micro-data restricted to CIU Rev. 4 Section J (Divisions 58–63); Banrep TRM. **Frequency:** monthly. **Window:** 2015-01-31 through 2026-03-31. **Sample:** $N = 134$ monthly. **Transformations:** logit on Section-J share; log-level on FX; same lag panel as Pair D; HAC($L = 12$); FEX_C_2018; empalme residual-bias diagnostic at the 2020-12 $\rightarrow$ 2021-01 Marco-2018 boundary. **Repo:** [notebooks/dev_ai_cost](#). **Spec sha:** 7c72292516...751f5a. **Result.** $\hat{\beta}_{\text{composite}} = -0.14613$, HAC SE 0.0847, $t = -1.726$, one-sided $p \approx 0.958$ against $H_1 : \beta > 0$; R-classifier MIXED (3 of 4 R-rows agree on negative sign). **Interpretation.** FAIL — sign-flipped at Section-J narrow cut. **Why it fails.** The Section-J substrate does not exhibit the lagged-FX-pass-through-to-employment pattern the BPO-offshoring narrative pre-supposed; an empalme-residual bias surfaced (the Phase-1 boundary diagnostic returned a $+0.375$ logit- Y jump at the 2020-12 $\rightarrow$ 2021-01 Marco-2018 boundary against a ± 0.335 envelope), so the post-2021 level shift is not fully neutralized at the Section-J cut. The R2 Section-M sensitivity arm (CIU Sections 69–75) returned $\hat{\beta}_{\text{composite}} = +0.45482801$, $t = +4.73$, $p = 1.13 \times 10^{-6}$, but this is a not-pre-registered candidate-next-iteration object, not a graduated PASS, and the rejection at Section-J falsifies the per-user instrument design that the dev-AI-paying population would have hosted.

FX-vol-CPI surprise — Colombian CPI through TRM realised volatility (FAIL). Colombian corporates and remittance recipients with FX-volatility exposure on COP/USD around DANE CPI publication, weekly cadence on a Friday anchor, with Y_w the weekly realised volatility of the Banrep TRM daily log-return panel and the regressor the lagged signed AR(1) CPI surprise.

$$\text{RV}_w^{\text{COP/USD}} = \alpha + \beta \text{Surp}_{w-1}^{\text{CPI}} + \varepsilon_w$$

where RV_w is realised volatility in week w and $\text{Surp}_{w-1}^{\text{CPI}}$ is the lagged signed monthly CPI surprise mapped to weekly cadence (carried across the release week, zero otherwise), with a small set of macro-surprise controls in the primary. **Data. Source:** Banrep TRM daily log-return panel; DANE CPI release calendar + realized values for AR(1) surprise construction. **Frequency:** weekly (Friday anchor). **Window:** 2008-01-02 through 2026-02-23. **Sample:** $N = 947$ weekly. **Transformations:** weekly RV from daily log-returns; signed AR(1) CPI surprise mapped to weekly cadence (carried across release week, zero otherwise); HAC(4) Newey-West SE; small set of macro-surprise controls. **Repo:** [notebooks/fx_vol_cpi_surprise](#). **Spec fingerprint:** nb1_panel_fingerprint.json. **Result.** $\hat{\beta}_{\text{CPI}} = -0.000685$, HAC(4) SE 0.001794, 90% interval $[-0.003636, +0.002266]$ contains zero; gate criterion $\hat{\beta} - 1.28 \cdot \widehat{\text{SE}} = -0.002981 < 0$ fires the FAIL. **Interpretation.** FAIL — no weekly-RV response to CPI surprises. **Why it fails and sensitivity caveat.** The pre-registered T1 exogeneity test rejects at $F = 15.12$, $p \approx 1.3 \times 10^{-9}$, so the iteration is a predictive regression rather than a structural-causal one and $\hat{\beta}$ cannot be read as an

impulse-response. Two pre-registered sensitivities returned positive significant 90% intervals at the one-sided $\alpha = 0.05$ convention — A1 monthly cadence $\hat{\beta} = +0.0152$ on $[+0.0057, +0.0246]$; A4 release-day-excluded $\hat{\beta} = +0.0033$ on $[+0.0005, +0.0062]$ — but neither is the iteration’s main finding: the pre-registered weekly primary is the result, and A1/A4 sit as candidate inputs to a possible future monthly-cadence-primary specification rather than as a rescue of the weekly primary.

Phase-A.0 Remittance — Colombian remittance corridor through FX (EXIT_NON_REMITTAN

Colombian household remittance recipients indexed against the Colombian remittance corridor with FX pass-through, with Y_w inherited from the FX-vol weekly panel and X_{w-1} a candidate on-chain remittance-flow proxy constructed from cCOP and COPm aggregate user activity on Celo.

$$RV_w^{\text{COP/USD}} = \alpha + \beta \Delta \text{Remit}_{w-1}^{\text{Col}} + \varepsilon_w$$

where $\Delta \text{Remit}_{w-1}^{\text{Col}}$ is the lagged Colombian household-remittance flow change as proxied on Celo on-chain activity, partitioned to attempt to isolate household remittance from third-party-DEX activity, treasury roundtripping, and bot arbitrage. *Data. Source (planned):* Ban-rep monthly remittance + Celo on-chain cCOP / COPm aggregate user activity. *Frequency:* weekly (planned; never reached final N). *Window:* 2024-09 through 2026-04 (planned). *Sample:* not produced — exited pre-estimation. *Transformations:* pre-registered partition rule for household remittance vs third-party-DEX activity, treasury roundtripping, bot arbitrage; never executed. *Spec:* [docs/specs/2026-04-20-remittance-surprise-trm-rv-design.md](#) (no notebook directory — spec exited pre-estimation). *Spec chain:* Phase-A.0 Rev 4.1 (kill-criterion-bearing). *Result.* No empirical estimate produced — exited on kill criteria pre-estimation 2026-04-24. *Interpretation.* EXIT — never estimated; killed by k1 + partial-k2. *Why it exits.* Task 11.F Axis-1 event research returned zero of thirty peak-day flows fingerprinting as remittance, with approximately 87% of surveyed flow decomposing into TRM arbitrage, treasury roundtripping, campaigns, and bot activity; the cCOP/COPM aggregate user activity on Celo is third-party-DEX volume, not Colombian household remittance, so no post-hoc filter could reconstruct an X-driver that was never measured upstream. The exit is load-bearing on the kill-criterion discipline: absent the pre-registered EXIT, the rev-4 plan’s non-stop filter iteration would have run to a filter that appeared to “rescue” a signal that was never captured.

P1 Bittensor SN18 — event-study apparatus (PARKED). The proprietary-AI policy-shock-exposed population (LATAM developers paying USD-denominated AI APIs as the downstream candidate; the apparatus pre-dates target-population narrowing), with Y_t a Bittensor SN18 (Cortex.t) alpha event-study indicator and X_t a set of pre-pinned policy-event dates (Bittensor halving 2025-12-15; dTAO mainnet activation 2025-02-13; Cortex.t milestones).

$$\text{AR}_{i,\tau} = \alpha_i + \sum_{\tau=-L}^{+L} \gamma_\tau \mathbf{1}\{t_i + \tau \in \mathcal{E}\} + \varepsilon_{i,\tau}$$

where $\text{AR}_{i,\tau}$ is the event-window abnormal alpha indicator at relative time τ for event i , $\mathcal{E}$ is the pre-pinned policy-event set, and the apparatus is evaluated over a nine-cell verdict cube (verdict-

eligibility crossed with robustness-consistency), Bonferroni-corrected at $\alpha_{\text{primary}} = 0.0167$ over the conjunctive family P1+P2+P3 with $N_{\text{min}}^{\text{events}} = 8$ and an asymmetric placebo gate. **Data.** **Source:** Bittensor SN18 (Cortex.t) alpha event-study; pre-pinned policy-event dates (Bittensor halving 2025-12-15; dTAO mainnet 2025-02-13; Cortex.t milestones). **Frequency:** event-window (asymmetric L). **Window:** per-event $\pm L$. **Sample:** planned $N_{\text{min}}^{\text{events}} = 8$; data ingest deferred. **Transformations:** event-window abnormal-alpha indicator; nine-cell verdict cube; Bonferroni at $\alpha_{\text{primary}} = 0.0167$ over P1+P2+P3. **Spec:** [docs/specs/2026-04-27-p1-sn18-event-study-design.md](#) (no notebook directory — apparatus parked pre-ingest). **Spec sha:** f855e036d3...b47aeab. **Result.** None — apparatus parked for the record on 2026-04-27; no graduating empirical claim produced. **Interpretation.** PARKED — apparatus complete, no claim graduated. **Why parked.** Apparatus complete; no graduating empirical claim. The user pivoted to the simple-composite- β fast-lane track (Pair D, dev-AI Stage-1) and parked P1 to preserve the intellectual capital of the spec work, with re-activation conditions documented in the parking memory.

§3. Risk-target framing

The risk target is FX-volatility exposure for a specific population: Latin-American digital workers paid in Colombian pesos (COP cash-flows) but carrying recurring USD-denominated professional-input obligations. When the COP weakens against the USD, the USD-denominated cost line translates into an unbudgeted COP-expenditure increase. The intermediation product target is a contract that compensates for this FX-pass-through cost shock, denominated in a unit the worker actually spends (COP, or a COP-pegged purchasing-parity stablecoin). The Abrigo Lemma's abstract $Y_{\text{inequality}}(t) = R_a(t) - R_c(t)$ framing — present in the 37-page technical companion §3 — is the eventual analytical generalisation; the entry-point letter introduces the concrete target only.

§4. Asks

§4.1 Suggested specifications

We invite the supervisor to suggest the (Y, X) pair, lag structure, sample window, primary inference convention, and robustness arms for the next iteration. Specifically: which sectoral cut should follow the Pair D / dev-AI G-T / J split given the R2 Section M signal; whether to repeat the FX-vol-on-CPI-surprise primary at A1 monthly cadence as the next pre-registration; or whether to pivot to a non-Colombian LATAM panel.

§4.2 Risks

The bullets below are risks the project has already identified; we ask the supervisor to add risks we have not flagged.

- Identification posture — predictive-regression versus structural-causal; T1 exogeneity rejected on FX-vol; HAC fixed- L per iteration.

- Convex-payoff sufficiency under insurance-underwriting standard — mean- $\hat{\beta}$ is necessary but not sufficient; [Carter et al. \(2017\)](#), [Mahul and Stutley \(2010\)](#), and [Clarke \(2016\)](#) set the standards.
- Multi- Y differential calibration — joint-leg correlation, FWER versus FDR, regime-switching.
- Anti-fishing protocol defensibility — $N_{\text{MIN}} = 75$, $\text{POWER}_{\text{MIN}} = 0.80$, $\text{MDES}_{\text{SD}} = 0.40$ inherited from Phase-A.0 Rev-5.3.1, not re-derived.
- Cross-iteration FWER and iteration-selection bias — five closed iterations; meta-multiple-comparison concern; [Heckman \(1979\)](#) as candidate frame.
- Stambaugh small-sample bias on persistent-regressor predictive regressions — [Stambaugh \(1999\)](#), [Phillips \(2014\)](#); log-FX is near-unit-root at monthly frequency.
- Empalme residual bias — DANE Marco-2018 methodology break not fully neutralised; surfaced in dev-AI Stage-1 NB02 Trio 1 boundary anomaly.

§4.3 Bibliography

The current bibliography is documented in the 37-page companion’s Appendix B and is reproduced as `refs.bib`. We invite the supervisor to recommend additional references he considers load-bearing for the framework’s empirical work, particularly on (a) parametric-index insurance pricing in low-counterparty-credit-standing populations, (b) sectoral employment-share predictive regressions on real-effective-exchange-rate panels, and (c) cross-iteration false-discovery-rate correction regimes for observational pre-registration.

§4.4 Connections

We invite the supervisor to introduce the project to other researchers, practitioners, or institutions whose work might complement this iteration roster — including but not limited to parametric-insurance underwriters with LATAM exposure, FX-derivatives operators serving sub-corporate notionals, on-chain settlement engineers, and applied econometricians working on premature deindustrialization in Latin America. The framework’s empirical work is open to collaboration; the supervisor’s network is the most efficient channel for identifying complementary contributors.

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